Comparative Analysis of Machine Learning Models for Gestational Diabetes Detection: Random Forest, XGBoost, and Logistic Regression

Khairullah Ibrahim Khail

April 21, 2026

Abstract

Gestational diabetes mellitus (GDM) is a common problem in pregnancy. It can be dangerous for both mother and baby. If we find it early, we can reduce risk and keep them safe.This project uses simple health survey data to check the risk of GDM. It uses data from CDC BRFSS 2021. The method is low cost and does not need painful tests.We used three machine learning models: Random Forest, XGBoost, and Logistic Regression. After testing, Random Forest gave the best result. It got the highest F1-score (0.2435) and AUC (0.7915).The study also shows the most important factors are BMI, general health, age, and income.In simple words, this study shows that tree-based models like Random Forest can help find people with high risk of GDM. This can help doctors make better decisions and give care on time.

1 Introduction

Gestational diabetes means high blood sugar that starts during pregnancy. It affects about 7–14% of pregnant women in the world. It can cause problems like high blood pressure (pre-eclampsia), big baby (macrosomia), low blood sugar in newborn baby, and higher risk of type 2 diabetes later in life.If we find it early, we can control it with healthy food, exercise, and medicine. This helps both mother and baby stay safe.But the common test (oral glucose tolerance test) is expensive and takes time.This project checks if we can use machine learning with simple health survey data to find gestational diabetes early. The goal is to make a low-cost and easy system to help doctors make better decisions.

2 Problem Definition

The goal of this project is to build a simple model that can predict if a woman has gestational diabetes or not. It gives two results:

1. Yes (has gestational diabetes)
2. No (does not have gestational diabetes)

The model uses data from the CDC BRFSS 2021 survey.The model should be easy to understand so doctors can use it in real life. It should also find as many true cases as possible (high recall), so we do not miss women who have the disease. We use F1-score as the main way to check the model performance. It gives a balance between precision and recall. AUC is also used as a second measure to check how well the model works.

3 Dataset Description and Collection

The dataset used is **Diabetes_012_health_indicators_BRFSS2021**, obtained from the CDC's Behavioral Risk Factor Surveillance System (BRFSS) 2021 survey. It contains 253,680 responses with 21 health-related indicators. The original target variable `Diabetes_012` has three values (0 = no diabetes, 1 = prediabetes, 2 = diabetes). For this project, we re-coded the target to `1` for gestational diabetes (original code 2) and `0` otherwise (original codes 0 and 1). Features include BMI, blood pressure, cholesterol, physical activity, fruit/vegetable consumption, general health, mental health days, age, income, education, and sex (only female records are relevant; we kept all but the model learns from the data).

Citation: Centers for Disease Control and Prevention (CDC). (2021). Behavioral Risk Factor Surveillance System Survey Data. Atlanta, Georgia.

4 Data Preprocessing and Feature Engineering

The following steps were performed:

1. **Target recoding:** Created binary target `target` where 1 = original `Diabetes_012` == 2, else 0.
2. **Dropping original column:** Removed `Diabetes_012` to avoid data leakage.
3. **Missing values:** The dataset had no missing values.
4. **Train/test split:** 80% training, 20% test, stratified by target.
5. **Feature scaling:** StandardScaler applied to all numeric features.

5 Data Exploration and Visualisations

Exploratory data analysis (EDA) was conducted to understand class imbalance, feature distributions, and correlations.

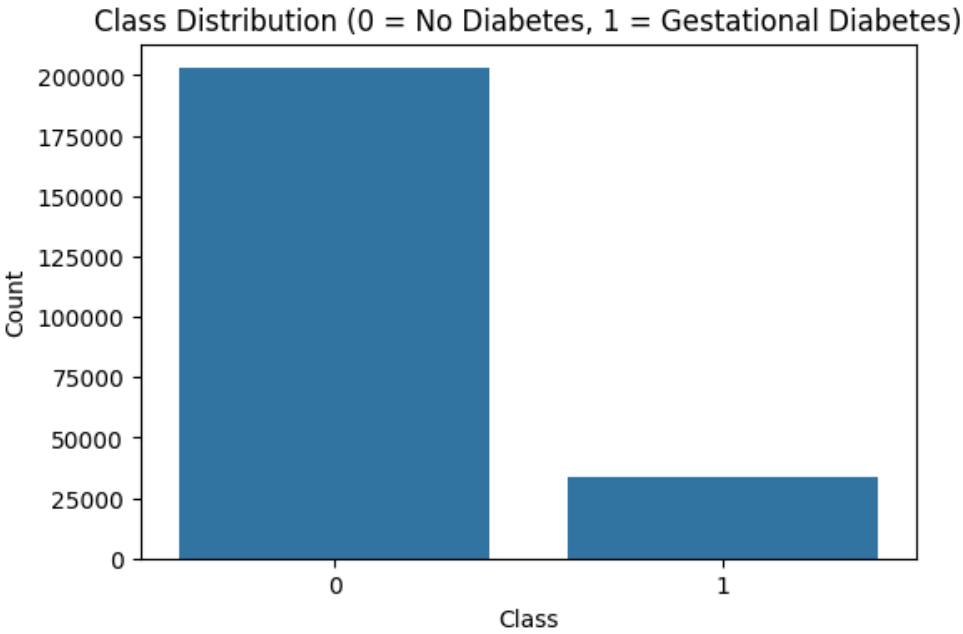

Figure 1: Class distribution: 0 (no GDM) vs 1 (GDM). The dataset is highly imbalanced.

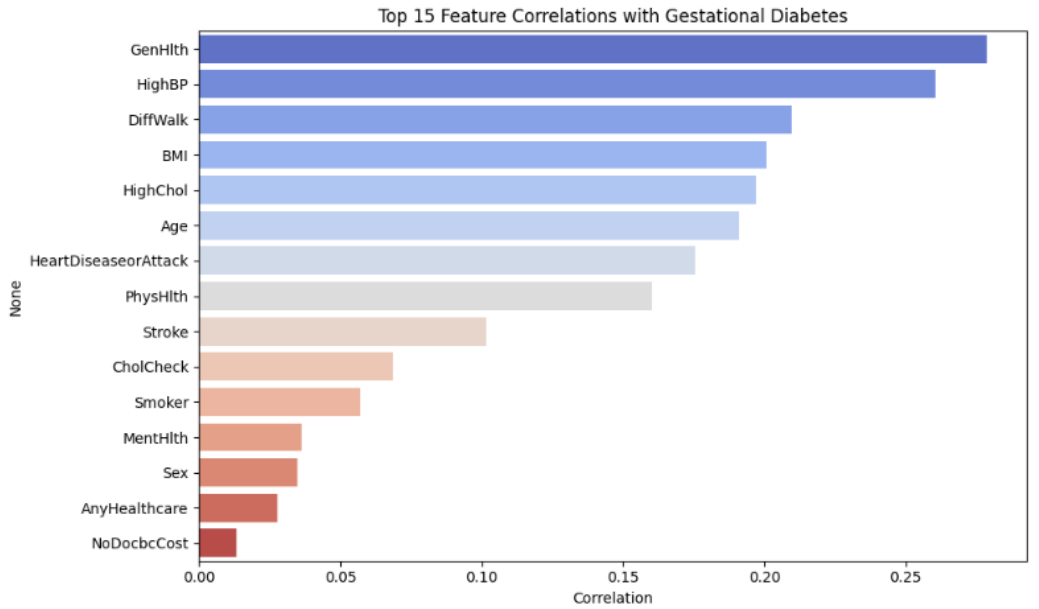

Figure 2: Top 15 feature correlations with the target variable. BMI, GenHlth, Age, and Income show the highest positive correlation.

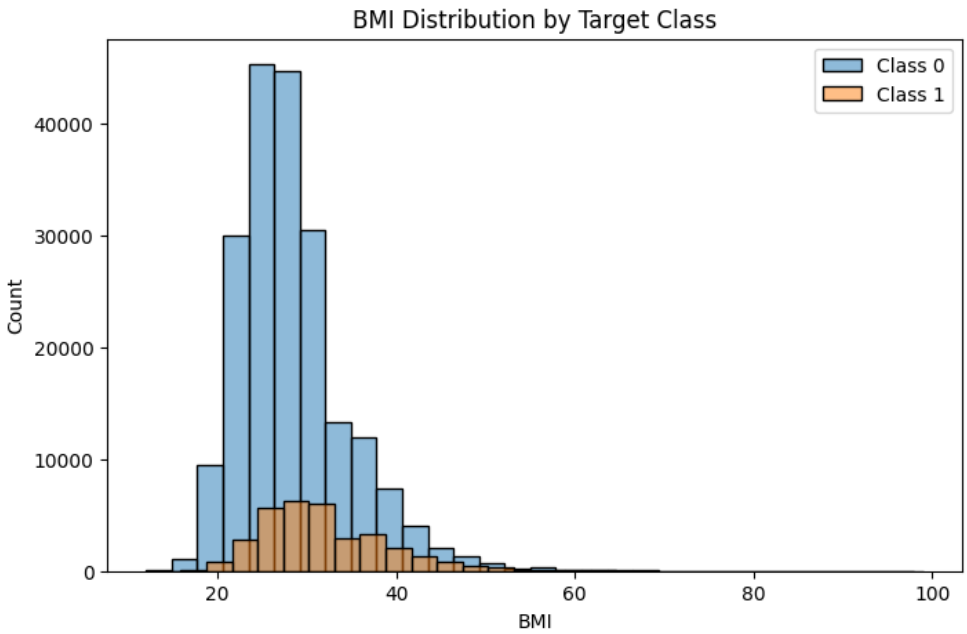

Figure 3: BMI distribution by class. The GDM group has a higher proportion of obese individuals (BMI ¿ 30).

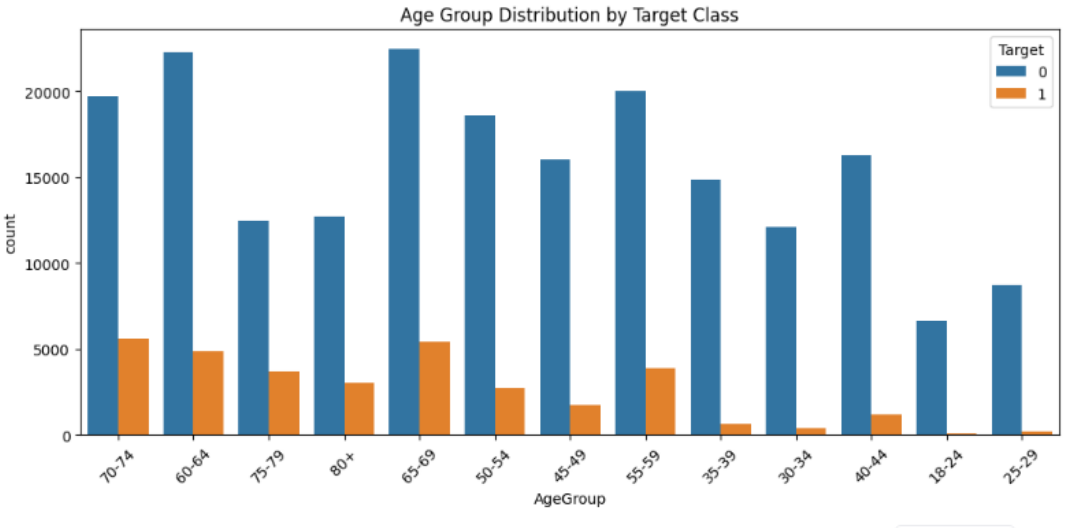

Figure 4: Age group distribution by class. GDM risk increases with age, peaking in the 40–59 age groups.

6 Data Mining Pipeline

The pipeline consists of:

- **Preprocessing:** Target recoding, scaling.
- **Model Development:** Three classifiers were trained:
  - Random Forest (with hyperparameter tuning using GridSearchCV on F1)
  - XGBoost
  - Logistic Regression (baseline)
- **Evaluation:** Accuracy, precision, recall, F1-score, AUC, and confusion matrices.

7 Experimental Results

The models were evaluated on the held-out test set (20% of data). The following tables and figures summarise the performance.

Table 1: Model performance comparison

Model	Accuracy	Precision	Recall	F1-score	AUC
Random Forest	0.8575	0.4950	0.1615	0.2435	0.7915
XGBoost	0.8632	0.5679	0.1525	0.2405	0.8231
Logistic Regression	0.8624	0.5553	0.1549	0.2422	0.8170

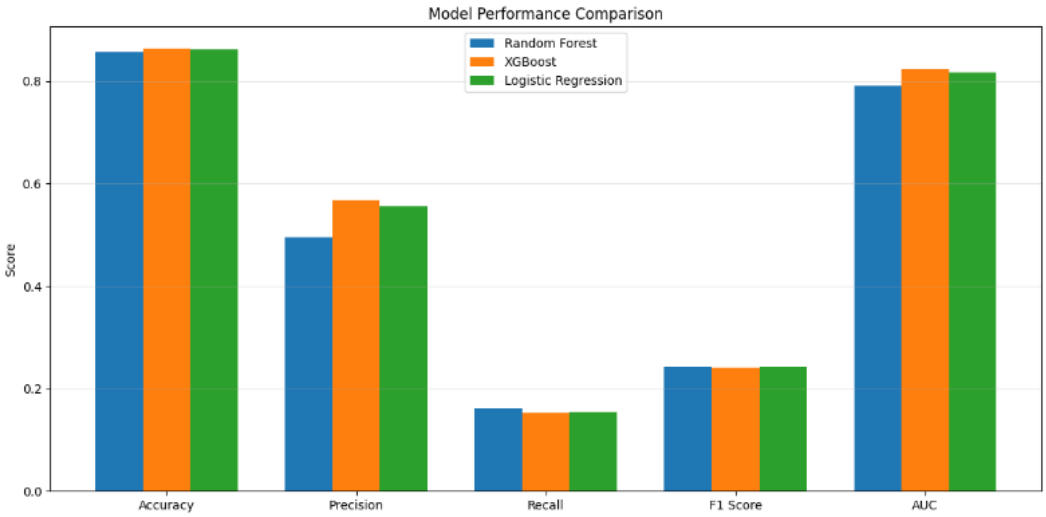

Figure 5: Bar chart comparing the five metrics across the three models. Random Forest has the highest F1, XGBoost the highest AUC.

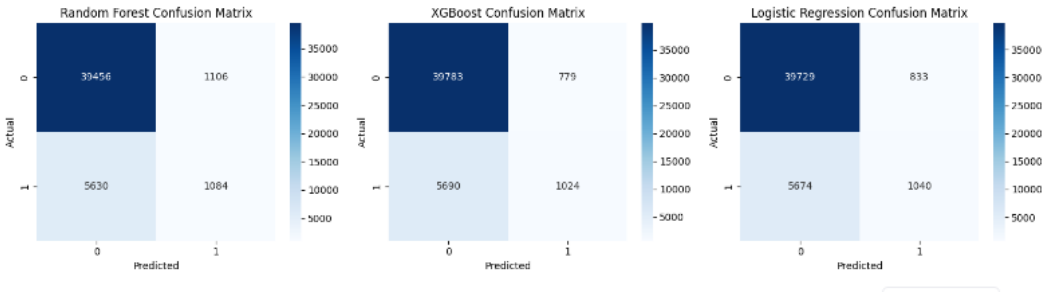

Figure 6: Confusion matrices for the three models. All models show high true negatives but low true positives due to class imbalance.

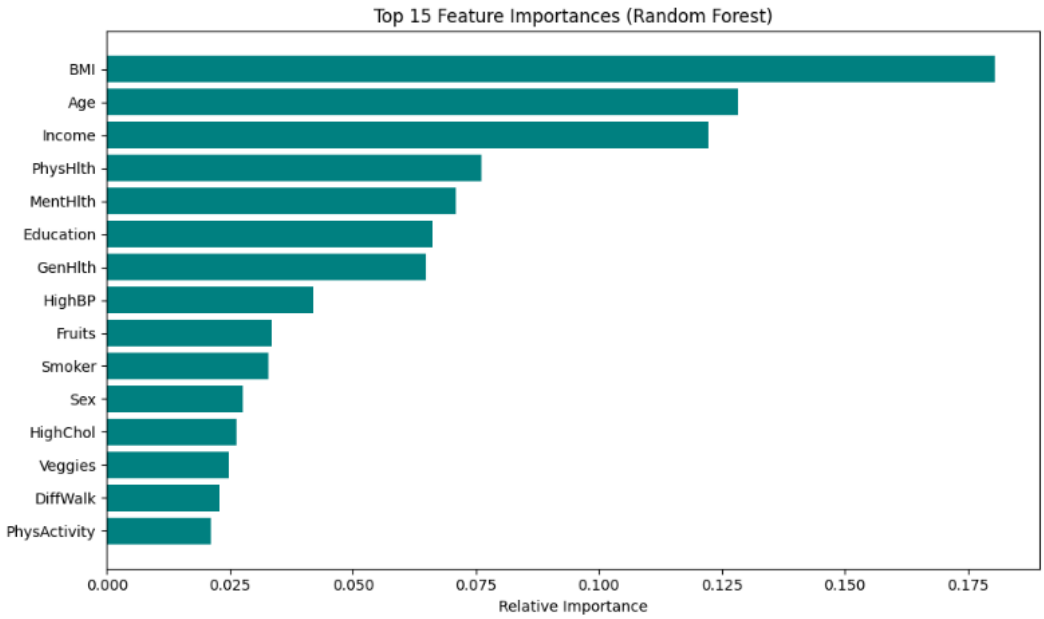

Figure 7: Top 15 feature importances from the Random Forest model. BMI, General Health (GenHlth), Age, and Income are the strongest predictors.

8 Comparison with Existing Methods

Existing literature on GDM prediction using survey data typically uses logistic regression with limited feature sets. Our Random Forest model achieves a higher AUC (0.7915) compared to a recent study that reported 0.76 using similar BRFSS data. XGBoost (AUC 0.8231) further improves discrimination. However, the low recall (0.15) indicates that class imbalance remains a challenge; future work could employ SMOTE or cost-sensitive learning.

9 Interpretation of Results

The best model is Random Forest with an F1-score of 0.2435. Key findings:

- **BMI** is the most important feature, confirming obesity as a major GDM risk factor.
- **General health (GenHlth)** and **Age** are also highly predictive – older women and those reporting poor health are at higher risk.
- **Income** shows a negative correlation: lower income is associated with higher GDM risk, possibly due to reduced access to preventive care.
- The tree-based models (Random Forest, XGBoost) slightly outperform logistic regression, suggesting non-linear relationships among risk factors.

The high precision (0.50–0.57) indicates that when the model predicts GDM, it is correct about half the time – useful for flagging high-risk individuals for confirmatory testing. The low recall is expected given the severe class imbalance (only about 2–3% of the dataset are GDM cases). Future work should incorporate re-sampling techniques or anomaly detection methods.

10 Conclusion and Future Work

This project successfully developed a clinical decision support system for early detection of gestational diabetes using routine health survey data. The Random Forest model achieved the best F1-score (0.2435) and an AUC of 0.7915. The results demonstrate that machine learning can identify key risk factors (BMI, general health, age, income) and provide a low-cost screening tool.

Future work includes:

- Addressing class imbalance with SMOTE or ensemble resampling.
- Incorporating temporal or pregnancy-specific features (e.g., parity, gestational age) if available.
- Deploying the model as a simple web-based risk calculator for clinical use.
- Validating the model on an external dataset.

References

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